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1. Project Plan

1.1 Project Overview

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Added a dedicated **Project Deliverables** section to clearly identify the expected artifacts and outputs of the project.
- Added a clear **Assumptions** section under Project Overview to document the assumptions used during project planning and development.
- Improved the **Constraints** section to clearly document project limitations and development constraints.
- Clarified the project's **Purpose, Scope, and Objectives** to improve the overall structure and readability of the Project Overview.

Reason for Change

- Addressed TA feedback that the **Project Overview lacked clearly defined Deliverables and Assumptions.**
- Improved the clarity and completeness of the project planning information.

1.2 Project Schedule

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Updated the project schedule to include specific **Target Start Dates** and **Target Completion Dates** for all five sprints.
- Added the schedule for:
 - **Sprint 1 (PA1): Authentication & Onboarding**
 - **Sprint 2 (PA2): Self-Study Dashboard & Utilities**
 - **Sprint 3 (PA3): Virtual Study Room & Flashcard**
 - **Sprint 4 (PA4): AI Speaking Assistant**

- **Sprint 5 (PA5): System Testing, Bug Fixing & Final Demo**

- Updated the schedule for all five corresponding incremental builds:
 - **Build 1.0 — Baseline System**
 - **Build 2.0 — Self-Study Dashboard Integration**
 - **Build 3.0 — Virtual Study Room Integration**
 - **Build 4.0 — Release Candidate**
 - **Build 5.0 — Final Production**
- Added specific milestone dates based on the updated academic/project schedule.
- Updated the final PA5 completion date to **15 August 2026** based on the current project schedule.

Reason for Change

- Addressed TA feedback that the **Project Schedule did not provide specific dates for all five sprints**.
- Improved schedule clarity by connecting sprint milestones with their corresponding incremental builds.
- Incorporated the latest available schedule information at this stage of the project.

1.3 Project Deliverables

Authors: [Minh Thự] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thự]

Changes

- Added a detailed **Project Deliverables** table.
- Documented the target delivery date and intended audience/target for each incremental build.
- Added the following deliverables:
 - Build 1.0
 - Build 2.0
 - Build 3.0
 - Build 4.0 — Release Candidate
 - Build 5.0 — Final Production

Reason for Change

- Addressed TA feedback that the **Project Overview did not clearly define project deliverables**.
- Improved traceability between project phases, sprint outputs, and final project artifacts.

1.4 Assumptions and Constraints

Authors: [Minh Thự] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thự]

Changes

- Added an explicit **Assumptions** subsection.
- Documented assumptions related to:
 - Team composition and responsibilities.
 - Incremental build delivery.
 - Final sprint demonstration.
 - Cross-functional pairing.
 - Centralized project documentation.
 - AI service availability and fallback mechanisms.
 - Rule-based evaluation for initial diagnostic tests.
- Refined the **Constraints** subsection to clearly document:
 - Five-sprint project structure.
 - Sprint duration limitations.
 - MVP scope limitations.
 - Feature scope freezing.
 - Change-impact assessment requirements.
 - 20% schedule buffer.
 - API monitoring and usage limitations.
 - AI grading validation requirements.

Reason for Change

- Addressed the lack of clearly documented project assumptions.
- Improved the clarity of project constraints and planning boundaries.

1.5 Roles, Responsibilities, Performers, and Reviewers

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Updated the project organization section to clearly identify the five team members and their assigned roles:
 - **Phạm Minh Thư — PM/BA/UI-UX**
 - **Lê Kim Hằng — Backend Lead**
 - **Nguyễn Khánh Linh — Backend Developer**
 - **Nguyễn Kim Thiên Phước — Frontend Lead**
 - **Hồ Gia Phúc — Frontend Developer**
- Expanded role descriptions to provide clearer responsibilities for each team member.
- Updated task assignments to identify specific **Performers/Doers** and **Reviewers** where applicable.

- Clarified the responsibilities of team leads in reviewing relevant frontend and backend development tasks.
- Improved the mapping between project roles and individual development activities.

Reason for Change

- Addressed TA feedback that **Performers/Reviewers were not clearly identified as specific individuals in some sections.**
- Improved accountability and responsibility tracking for project tasks.

1.6 Management Process

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Expanded the Management Process section to provide clearer information about:
 - Project estimates.
 - Project planning.
 - Phase and sprint planning.
 - Incremental builds.
 - Project scheduling.
 - Project resourcing.
 - Project monitoring and control.
- Added clearer descriptions of project monitoring mechanisms through:
 - Sprint schedules.
 - Task assignments.
 - Review assignments.
 - Due dates.
 - Incremental builds.
 - Risk management.
 - Project reporting.
 - Testing and bug fixing activities.

Reason for Change

- Improved the completeness and clarity of the Software Development Plan.
- Better aligned the management process with the project's current development stage.

1.7 Requirements and Scope Management

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Clarified the MVP scope and its focus on:
 - The IT track.
 - Core CEFR proficiency tracking.
- Clarified the scope control mechanisms:
 - Formal change-impact assessment for new feature requests.
 - Dedicated "Nice-to-Have" backlog for Phase 2 features.
 - Feature scope freezing for each sprint.
- Improved the description of how requirements and scope are managed throughout the project.

Reason for Change

- Improved scope control and reduced the risk of scope creep.
- Clarified how new feature requests are handled during sprint execution.

1.8 Risk Management

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Updated the Risk Management section with identified project risks and corresponding mitigation strategies:
 - Scope Creep.
 - API Failures or Inaccurate CEFR Grading.
 - Member Unavailability & Bottlenecks.
- Added mitigation strategies involving:
 - Strict MVP definition.
 - Nice-to-Have backlog.
 - Change-impact assessment.
 - Sprint scope freezing.
 - Graceful degradation for unavailable AI services.
 - Rule-based evaluation.
 - API monitoring and billing alerts.
 - Daily usage caps.
 - Human validation of AI-generated CEFR scores.
 - Cross-functional pairing.
 - Centralized documentation.
 - 20% sprint milestone buffer.

Reason for Change

- Improved risk visibility and mitigation planning.
 - Updated risk management based on the project's current technical and organizational context.
-

1.9 Reporting and Measurement

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Clarified the project's reporting activities, including:
 - Weekly Reports.
 - AI Usage Reports.
 - Final Presentation Slides.
 - Product Demo.
 - PA5 Project Report.
- Added clearer task-level monitoring through:
 - Task Description.
 - Assignee/Doer.
 - Reviewer.
 - Due Date.
- Clarified the use of incremental builds to monitor project progress.

Reason for Change

- Improved project progress tracking and reporting clarity.
 - Provided clearer connections between project tasks, responsible members, reviewers, and deliverables.
-

1.10 Configuration and Scope Control

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Clarified project configuration and scope control practices:
 - Maintaining centralized project documentation.
 - Performing change-impact assessments.
 - Freezing sprint scope.
 - Maintaining a Nice-to-Have backlog.
 - Tracking project documentation and deliverables.

Reason for Change

- Improved control over project scope and project artifacts.
- Reduced the risk of uncontrolled changes during sprint execution.

Summary of Changes in Project Plan

The **Project Plan Version 2.0** was revised from **Version 1.0 (PA2-2026)** to address TA feedback and incorporate updated project information.

The major changes include:

1. Added a clearly defined **Project Deliverables** section.
2. Added an explicit **Assumptions** section.
3. Updated the **Project Schedule** with specific dates for all five sprints and their corresponding builds.
4. Added and updated target delivery dates for project deliverables.
5. Clarified **Performers/Doers and Reviewers** by identifying specific team members where applicable.
6. Expanded and clarified **Roles and Responsibilities**.
7. Improved the **Management Process** and project monitoring descriptions.
8. Clarified **Requirements and Scope Management**.
9. Updated and expanded **Risk Management** and mitigation strategies.
10. Improved **Reporting and Measurement** descriptions.
11. Clarified **Configuration and Scope Control** practices.
12. Updated project information based on the latest available schedule and development progress.

2. Vision Document

2.1 Applicable Standards

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Added a dedicated **Applicable Standards** section to the Vision Document as **Section 4.3**.
- Specified the **Common European Framework of Reference for Languages (CEFR)** as the primary standard used by Studify for English proficiency tracking and personalized learning paths.
- Clarified that the project focuses on the **IT track** and **core CEFR proficiency tracking** within the MVP scope.
- Clarified that the CEFR framework is used to support proficiency benchmarking and learning progression across the target users.

Reason for Change

- Addressed TA feedback that the **Vision Document was missing the Applicable Standards section**.

- Clearly identified the **CEFR framework** as the applicable standard for the English learning and proficiency-tracking aspects of Studify.
- Improved the clarity of the product's learning and assessment framework.

2.2 Maintainability Requirements

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Added a dedicated **Maintainability Requirements** section as **Section 6.8** under Other Product Requirements (Non-Functional Requirements).
- Added a minimum automated **unit and integration test coverage requirement of 70%**.
- Added a requirement for **static code analysis and linting**, with zero critical or blocking errors before code merging.
- Added a **Documentation Currency** requirement to ensure that centralized project documentation, including API specifications, architecture diagrams, and onboarding guides, is updated within 24 hours of any major release.
- Added a **Knowledge Distribution** requirement to ensure that critical system components have a minimum **Bus Factor of 2**, supported by cross-functional pair programming and code reviews.
- Added **CI/CD requirements** for automated testing, incremental builds, and staging deployments.
- Defined a requirement for pull-request build and validation cycles to complete within **10 minutes**.

Reason for Change

- Addressed TA feedback that the Vision Document lacked **Maintainability Requirements**.
- Defined measurable requirements to ensure the codebase remains maintainable and that critical system knowledge is distributed across the development team.
- Improved long-term project sustainability through automated testing, documentation, code reviews, and CI/CD practices.

2.3 Scalability Requirements

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Added a dedicated **Scalability Requirements** section as **Section 6.9** under Other Product Requirements (Non-Functional Requirements).
- Defined a requirement for the system to support at least **200 concurrent active users** during sustained study, timer, and quiz sessions.
- Added a requirement to maintain peak application-server CPU usage below **75%** under the target concurrent load.

- Added measurable **performance-under-load requirements**, including:
 - Average system API response time below **2.0 seconds (p95 < 2.0s)** for general requests under normal operating load.
 - Static asset delivery response time below **500 ms**.
- Added **API rate limiting** of **60 requests per minute per user** for standard endpoints.
- Added strict **daily usage caps** for external AI/LLM API calls to control external service usage and costs.

Reason for Change

- Addressed TA feedback that the Vision Document lacked **Scalability Requirements**.
- Defined measurable system capacity and performance targets to ensure that Studify can support the expected number of concurrent users.
- Added rate limiting and AI API usage controls to reduce the risk of service overload, unexpected rate limits, and excessive external API costs.

2.4 Accessibility Requirements

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Added a dedicated **Accessibility Requirements** section as **Section 6.10** under Other Product Requirements (Non-Functional Requirements).
- Defined **WCAG 2.1 Level AA** as the accessibility compliance target for the Studify web interface.
- Added a requirement that key user interaction flows must be fully accessible through **keyboard navigation**.
- Specified keyboard accessibility for important interactions, including:
 - Flashcard navigation.
 - Flashcard flipping.
 - Pomodoro timer controls.
- Added visual accessibility requirements for **color contrast**, requiring:
 - A minimum contrast ratio of **4.5:1** for normal text.
 - A minimum contrast ratio of **3:1** for large text and UI components.
- Added a requirement to support **text scaling up to 200%** without loss of content or functional overlap.

Reason for Change

- Addressed TA feedback that the Vision Document lacked **Accessibility Requirements**.

- Established measurable accessibility criteria to make the Studify interface more usable for a wider range of users.
- Improved consistency between the product's usability goals and accessibility expectations.

2.5 Data Quality Requirements

Authors: [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Gia Phúc, Khánh Linh] | **Editor:** [Minh Thư]

Changes

- Added a dedicated **Data Quality Requirements** section as **Section 6.11** under Other Product Requirements (Non-Functional Requirements).
- Added requirements for **AI Grading Accuracy and Validation**, requiring AI-generated CEFR scores to be benchmarked against human-evaluated control samples.
- Defined a minimum **85% score alignment rate** or alignment within a **±1 sublevel threshold** for AI-generated CEFR assessments.
- Added a **Deterministic Baseline Evaluation** requirement for diagnostic placement tests, ensuring that static multiple-choice components use pre-tested rule-based evaluation mechanisms.
- Added a requirement for **Proficiency Standard Uniformity**, ensuring that CEFR definitions from **A1 through C2** are consistently applied across progress tracking and score aggregation.
- Added a **Data Recovery and Loss Limits** requirement with:
 - **RPO ≤ 1 hour.**
 - **RTO ≤ 2 hours.**
- Applied data quality requirements to important user data, including study history, flashcard decks, progress metrics, and proficiency scores.

Reason for Change

- Addressed TA feedback that the Vision Document lacked **Data Quality Requirements**.
- Established measurable criteria for the accuracy, consistency, standardization, and recoverability of critical learning data.
- Improved the reliability of AI-generated CEFR assessments and user progress tracking.
- Reduced the potential impact of data corruption or system failures on user learning history and progress information.

Summary of Changes in Vision Document

The **Vision Document Version 2.0** was revised from **Version 1.0** to address TA feedback and improve the completeness of the product requirements.

The major changes include:

1. Added a dedicated **Applicable Standards** section identifying **CEFR** as the primary standard for English proficiency tracking and personalized learning paths.
2. Added **Maintainability Requirements** covering automated test coverage, static code analysis, documentation currency, knowledge distribution, and CI/CD practices.
3. Added **Scalability Requirements** covering concurrent user capacity, performance under load, CPU usage, API rate limiting, and external AI/LLM usage caps.
4. Added **Accessibility Requirements** covering WCAG 2.1 Level AA compliance, keyboard navigation, visual contrast, and text scaling.
5. Added **Data Quality Requirements** covering AI grading accuracy, deterministic diagnostic evaluation, CEFR standard uniformity, and data recovery objectives.
6. Added measurable and verifiable non-functional requirements to improve the clarity and completeness of the Vision Document.
7. Expanded Section 6 from the original set of **Performance, Security, Platform, Reliability and Availability, and Usability Requirements** to include additional requirements for **Maintainability, Scalability, Accessibility, and Data Quality**.
8. Improved the alignment between the Vision Document and the TA feedback received for the previous submission.